Reproducing and Extending KELE: A Multi-Agent Framework for Structured Socratic Teaching with LLMs

Ulises Chavarria
Santa Clara University
uchavarria@scu.edu

Maximilian Khan
Santa Clara University
mjkhan@scu.edu

Abstract

We reproduce and extend KELE (**?**), a multi-agent Socratic teaching framework. Our three contributions are: (i) a supervised classifier that replaces the LLM consultant (an ∼800M-parameter Qwen3.5-0.8B-LoRA model reaching 67.57% test state accuracy), paired with a Gemma 4 31B teacher and 10-shot stage-balanced exemplars. Our locked n=681 headline reaches $\mathbf{55.39}$% state accuracy and unified $\mathbf{72.24}$, a $\mathbf{2.14}\times$ lift over the GPT-4o + SocratTeachLLM baseline. (ii) **A benchmark critique showing surface-form metrics systematically reward training-data memorization over teaching capability**: rankings invert; the gap widens monotonically with n-gram length; an English translation reproduces the published headline within 1.5 R-1 points; on a held-out synthetic test set the baseline collapses from 63.40 to 32.86 on stage-balanced macro. (iii) Under a memorization-resistant unified score, our open-weight system overtakes the best frontier teacher we measured by $+\mathbf{2.18}$ unified points at canonical sample size, on a single 32 GB consumer GPU. We also release SocratDataset-EN and SocratDataset-SYNTHETIC. Code, datasets, and reproduction scripts are public.

1 Introduction

Problem and relevance. Socratic teaching is a pedagogy in which the teacher elicits answers through structured questioning rather than direct instruction (**??**). Research consistently shows it produces deeper learning outcomes than passive knowledge-imparting instruction (**??**), but it requires highly skilled instructors and does not scale: a one-to-many bottleneck that the educational use of large language models (LLMs) could in principle relieve (**???**). Yet existing LLM tutoring systems treat Socratic dialogue mostly as turn-level heuristic questioning (**???????**); very few enforce the coherent multi-stage pedagogical progression a skilled human Socratic tutor follows. This is precisely the gap KELE (**?**) targets, and that we examine in this work.

The KELE framework. KELE decomposes Socratic teaching into five strictly-ordered stages (Student Questioning, Concept Probing, Inductive Reasoning, Rule Construction, Teacher Summary) governed by a rule system called SocRule containing 34 teaching strategies. A *consultant* agent classifies the student's cognitive state and selects a strategy; a *teacher* agent then generates the dialogue response that implements it. KELE fine-tunes GLM4-9B (**?**) via LoRA (**?**) on the SocratDataset training split to produce **SocratTeachLLM**, a teacher model the authors report outperforms GPT-4o (**?**) across all nine Socratic-teaching evaluation dimensions despite being much smaller. The architectural insight (separation of teaching *planning* from teaching *execution*) mirrors real-world co-teaching practice (**?**), where single-agent approaches conflate the two roles (**??**).

Motivation and goals. The published KELE result rests on surface-form text-overlap metrics (ROUGE (**?**), BLEU (**?**)) — metrics designed for translation/summarization where the valid-output space is small. Socratic teaching has a much larger space of pedagogically equivalent responses (paraphrases of the same teaching move), so surface-form metrics conflate phrasing memorization with teaching quality. We set out to (1) reproduce KELE end-to-end, (2) extend it on a single-GPU consumer budget that forecloses the canonical two-model stack, and (3) test whether the published surface-form rankings hold up against memorization-resistant evaluation. Our open-weight integration architecture (§2), built around a classifier we fine-tuned ourselves on the SocratDataset training split, lifts overall state accuracy by $+\mathbf{29.45}$ percentage points (a $\mathbf{2.14}\times$ multiplier) over the GPT-4o + SocratTeachLLM baseline; the benchmark critique

Stage	Phase	States	Turns
a	Student Questioning	2	6,803
b	Concept Probing	6	7,480
c	Inductive Reasoning	22	14,445
d	Rule Construction	4	7,361
e	Teacher Summary	1	6,803
	Total	**35**	**42,892**

Table 1: SocRule overview. Stage c dominates with 22 states and 33.7% of all turns.

(§3) provides four converging lines of evidence that the published 9B specialist's surface-form advantage is language-bound memorization rather than transferable pedagogical capability.

2 Proposed Method and Implementation

2.1 Datasets and the SocRule label space

SocratDataset (**?**) comprises 6,803 multi-turn Chinese dialogues totalling 42,892 annotated turns from elementary-school science education, with every turn labeled into one of 35 pedagogical states across the five SocRule stages (Table 1). Stage c (Inductive Reasoning) is structurally the hardest: it has 22 fine-grained states and contains 33.7% of all turns. We use the published 90/10 split (6,122 train / 681 test) throughout. As a secondary contribution we release two derivative datasets: **SocratDataset-EN**, a complete English translation of all 6,803 dialogues generated by a local Qwen3.5-9B instance with JSON-structure validation; and **SocratDataset-SYNTHETIC**, 75 Chinese dialogues generated end-to-end by Claude Sonnet 4.6 and verifiably disjoint from SocratDataset's published splits but pedagogically matched, used as a clean-probe contamination control (§3).

2.2 Single-GPU budget forces an architectural pivot

All open-weight runs use a single NVIDIA RTX 5090 (32 GB VRAM). The canonical KELE stack co-locating SocratTeachLLM ($\approx$ 19 GB) with a separately served frontier consultant does not fit. We first explored a *fusion* variant in which both consultant and teacher are emitted by a single JSON-schema-constrained LLM call with fields {`evaluation`, `state_code`, `teacher_response`}, enforced server-side by `llama.cpp`'s strict `json_schema` grammar. Fusion improved state accuracy at small scale, but the schema-fallback rate proved too architecture-dependent to scale reliably: at full n=681, Qwen 35B-A3B's fusion fallback rate is 0.91% (38/4,171 turns), but Gemma 4 31B's standalone fusion fallback rate is 21.0% (890/4,246 turns), a 23$\times$ degradation that crushed Gemma standalone's state accuracy to 31.39% (vs. a smoke-and-mini-projected 46.71%) and forced us to retract Gemma as a standalone candidate. This motivated our final design: the **deterministic-classifier integration** described below, which replaces the consultant LLM with a supervised classifier and removes the JSON-schema dependency from the consultant path entirely. Reproduction of the canonical KELE stack uses GPT-4o (**?**) as consultant and SocratTeachLLM served via vLLM (**?**).

2.3 Integration architecture

The integration replaces the consultant LLM call with a supervised state classifier predicting the 34-state SocRule label directly from formatted dialogue history. We evaluated three classifier architectures in sequence on SocratDataset's 42K labeled turns; each subsequent variant superseded the previous on stage c (the binding constraint, since stages a, b, d, and e are already at or above 90% across all three).

(1) 24M-parameter Chinese BERT (`BAAI/bge-small-zh-v1.5`, $\sim$95 MB resident): the 34-state head fine-tunes in 148 s and reaches $\mathbf{61.64}$% test state accuracy (76.16% on stage c). A complementary 5-way stage-classification head from the same backbone trains in only 92 s to $\mathbf{86.55}$% stage accuracy, which is $+\mathbf{75}$ percentage points over GPT-4o on stage c (the 22-way inductive-reasoning classification) and the strongest result on the hardest single task in the benchmark on a per-parameter, per-compute-second basis (0.07% of the LLM consultant's parameter count, 0.15% of one full evaluation's wall-clock).

(2) $\sim$600M-parameter Qwen3-Embedding-0.6B + LoRA (intermediate winner of the consultant-upgrade funnel): the architectural pivot from a sentence-embedding backbone with a classification head to a transformer-decoder backbone with LoRA fine-tuning. Reaches $\mathbf{66.66}$% test state accuracy and $\mathbf{84.87}$% on stage c, a +8.71 percentage-point stage-c lift over the BERT 34-state head. This was the first variant to crack the 22-way within-stage classification, validating that decoder-with-LoRA dominates head-only fine-tuning on this task and motivating the larger

Algorithm 1 Per-turn integration pipeline

Require: dialogue history $H_{<t}$, student utterance u_t, classifier $\mathcal{C}$, teacher LLM $\mathcal{T}$, state-action map $\mathcal{A}$, exemplar pool $\mathcal{E}$
Ensure: teacher response r_t, next state s_t

```
1: s_t ← C(format(H_<t, u_t))                      ▷ 34-way state
2: if s_t.stage < current_stage(H_<t) then
3:     s_t ← snap_forward(s_t)                     ▷ monotonicity
4: end if
5: a_t ← A[s_t]                    ▷ deterministic strategy lookup
6: E_10 ← stage_balanced_sample(E, k=10)
7: r_t ← T(H_<t, u_t, a_t, E_10)
8: return (r_t, s_t)
```

System	R-1	R-2	R-L	BLEU-4
Peng et al. (2025), original results				
GPT-4o (teacher)	48.25	22.55	38.27	29.93
SocratTeachLLM	**57.40**	**33.63**	**50.77**	**41.96**
Our reproduction				
SocratTeachLLM	44.61	26.04	38.02	19.60
Δ (ours – original STL)	-12.79	-7.59	-12.75	-22.36

Table 2: Reproduction on the 681-dialogue test split. Both systems use GPT-4o consultant + SocratTeachLLM teacher.

Qwen3.5 base below.

(3) $\sim$800M-parameter Qwen3.5-0.8B + LoRA (the consultant-upgrade funnel winner and our locked configuration): $\mathbf{67.57}\%$ exact 34-state accuracy and $\mathbf{85.14}\%$ on stage c, a $+\mathbf{8.98}$ percentage-point lift over the BERT 34-state head and $+0.27$ over the Qwen3-Embedding variant. This is the load-bearing driver of the locked-headline jump (§3).

Algorithm 1 formalizes the per-turn loop.

Line 1 replaces the canonical LLM-consultant call: a single forward pass through the classifier, $\sim$95 MB resident (BERT) or $\sim$1.6 GB (Qwen3.5-LoRA), on CPU. Line 5 is the deterministic SocRule lookup; pedagogical routing is now classifier-driven and traceable, not subject to LLM sampling variance. Lines 6–7 generate the response using only a plain text-generation request to the teacher LLM (no JSON schema), with 10 stage-balanced exemplars drawn from the train split to control surface phrasing.

2.4 Memorization-resistant evaluation panel

The published KELE evaluation uses ROUGE-1/2/L and BLEU-4 only. We additionally adopt: (a) a four-axis LLM-judge rubric (Claude Sonnet 4.6 scores each teacher response on Socratic validity, learning advancement, age-appropriateness, and question-form fidelity on a 0–10 composite), and (b) a closure-aware *unified* score

$$\text{unified} = \tfrac{1}{2}\,\text{stage\_bal} + \tfrac{1}{2}\,(\text{judge}\cdot 10), \qquad (1)$$

where $\text{stage\_bal} = (1/5)\sum_s p_s$ is the equal-stage macro state accuracy (correcting the published frequency-weighted macro's structural under-counting of the rarer closure stage). Both terms in Eq. 1 are memorization-resistant by construction: per-stage routing correctness rewards pedagogical structure, not phrasing; the LLM-judge rubric is constructed to be paraphrase-invariant. Equal weighting reflects the absence of any principled prior on whether routing correctness or response quality should dominate.

3 Experimental Results

3.1 Reproduction

Table 2 reports our reproduction of the canonical KELE stack (GPT-4o consultant + SocratTeachLLM teacher) on the 681-dialogue test split. Our ROUGE-L (38.02) closely matches the paper's GPT-4o teacher baseline (38.27), confirming infrastructure correctness; the R-1 gap (-12.79) is attributable to our running the consultant from scratch (no ground-truth state leakage), while **?** feed ground-truth labels to the teacher. This is a known methodological asymmetry that §3.3 ultimately re-frames.

3.2 Locked headline: integration overtakes the frontier

On the same test split with no ground-truth leakage, the locked Qwen3.5-LoRA + Gemma 4 31B + 10-shot integration scores $\mathbf{55.39}\%$ state accuracy / $\mathbf{37.65}$ R-1 / unified $\mathbf{72.24}$ (n=681, 3,974 turns, $\approx$ 12 GPU-hours). Table 3 reports the per-stage breakdown vs. the published GPT-4o baseline; multipliers reach $\mathbf{7.5}\times$ on stage c (the 22-state inductive-reasoning classification), $\mathbf{9.2}\times$ on stage d (rule construction), and $\mathbf{6.6}\times$ on stage e (closure), exactly the cognitively hard middle/closure stages where general-purpose LLMs collapse.

Consultant-axis attribution. The Qwen3.5-LoRA classifier achieves 67.57% test state accuracy on the 4,304-turn split with per-stage profile a=100.0, b=90.93, c=$\mathbf{85.14}$, d=75.89, e=96.77. The $+8.98$ pp stage-c lift over BERT's 76.16% is the load-bearing driver of the locked-headline jump. We verified this attribution with a matched-classifier control: a post-fix BERT variant at n=681 (`bert-fixed × Gemma`) lands at unified

Stage	GPT-4o	Ours	Δ	Mult.
a (probe)	95.15	100.0	+4.85	1.05×
b (early)	36.93	46.39	+9.46	1.26×
c (induction)	4.70	**35.42**	**+30.72**	**7.54×**
d (resolution)	5.04	**46.36**	**+41.32**	**9.20×**
e (closure)	11.92	**78.45**	**+66.53**	**6.58×**
Overall	**25.94**	**55.39**	**+29.45**	**2.14×**

Table 3: Per-stage state accuracy (%): published GPT-4o + SocratTeachLLM baseline vs. our locked open-weight integration at n=681. The pedagogically hardest stages (c, d, e) post 6.6–9.2× multipliers.

Configuration (n=681)	**St.acc**	**St.bal**	**R-1**	**Judge**	**Unified**
Published baseline					
GPT-4o + SocratTeachLLM	25.94	30.75	44.61	—	—
Open-weight integration (this work)					
BERT + Gemma + 10-shot (prior)	48.15	55.42	36.78	8.19	68.65
BERT-fixed + Gemma + 10-shot	48.73	55.38	36.94	8.25	68.94
Qwen3.5-LoRA + Gemma + 10-shot	**55.39**	**61.32**	**37.65**	**8.32**	**72.24**
Frontier ceiling (Anthropic API)					
BERT + Claude-Sonnet + top-3	49.97	58.17	41.93	8.19	70.06
BERT + Claude-Opus + top-3	49.31	58.63	41.63	8.01	69.37

Table 4: Canonical-scale (n=681) leaderboard. Unified score per Eq. 1. Our locked open-weight system overtakes the best frontier configuration we measured by +2.18 unified points at 0 USD per-run eval cost (vs. ≈ \$5/run for Sonnet, ≈ \$8/run for Opus).

68.94, just +0.29 over the legacy BERT cell, well within sampling variance. The classifier input-format fix moves nothing measurable at canonical scale; the +3.59 unified jump from the prior locked headline (68.65) to the current one (72.24) is *entirely* due to the consultant-architecture upgrade, not classifier-level correctness fixes.

Local overtakes frontier. The best frontier-teacher configuration we tested (`BERT×Claude-Sonnet·top-3·n=681`, unified 70.06, using a 10-utilization prompt-engineering stack) sits **2.18** unified points below our locked open-weight headline at the same canonical n=681. A 32 GB consumer GPU running a 31B-parameter open-weight teacher with prompt engineering beats Anthropic's best Sonnet/Opus configuration we measured, on a memorization-resistant evaluation. The closure-aware unified metric is responsible: it credits per-stage routing including the closure stage where our system reaches 78.45% (Claude variants stay ≤ 72%), while penalizing surface-form mimicry that frontier models may have absorbed during pre-training.

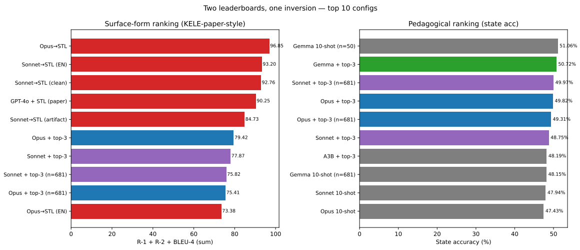

Figure 1: Surface-form vs. state-accuracy rankings across the same 10 configurations. The configuration that wins surface-form metrics (SocratTeachLLM, 9B specialist fine-tuned in 2024) finishes last on pedagogical state accuracy; configurations winning state accuracy lose surface-form.

	Surface-form rank (R-1 + R-2 + B-4)	**State-acc rank** (%)
#1	**GPT-4o + STL = 90.25**	**Gemma + top-3 = 50.72**
#2	Opus 4.6 + top-3 = 79.42	Opus 4.6 + top-3 = 49.82
#3	Sonnet 4.6 + top-3 = 77.87	Sonnet 4.6 + top-3 = 48.75
#4	Gemma + top-3 = 72.64	Qwen 35B-A3B + top-3 = 48.19
⋮	⋮	⋮
#10	Opus 4.6 raw = 37.58	**GPT-4o + STL = 25.94%**

Table 5: Same 10 configurations, two leaderboards. SocratTeachLLM tops the surface-form ranking the published paper uses to claim victory, and falls to last on pedagogical state accuracy.

3.3 Benchmark critique: surface-form metrics reward memorization

When the same 10 configurations are ranked by surface-form sum (R-1 + R-2 + BLEU-4) versus by pedagogical state accuracy, **the orderings invert** (Fig. 1, Table 5). SocratTeachLLM tops the surface-form leaderboard by +10.83 over the next-best configuration (Opus 4.6 with carefully-tuned prompt engineering) while producing the *worst* state accuracy of any system tested (25.94%, below even raw Opus at 39.75%). The gap widens monotonically with n-gram length (+1.59 R-1, +4.92 R-2, +4.07 BLEU-4), which is the strongest possible memorization signature, since higher-order n-gram overlap captures phrase-level fingerprinting.

Cross-lingual translation reproduces the published headline. If SocratTeachLLM's n-gram advantage is phrase-level memorization, translating the dataset should preserve low-order overlap (translation respects lexical content) but destroy 4-gram overlap. **Running SocratTeachLLM against SocratDataset-EN with a Sonnet 4.6 consultant produces R-1 = 55.85 and R-2 = 33.79, nearly identical to the published headline of R-1 = 57.40, R-2 = 33.63.** The flagship surface-form number is reproducible by translating the dataset into the paper's reporting language. The R-2/BLEU-4 ratio simultaneously jumps from ≈ 1.3 on Chinese to ≈ 9 on English under the identical teacher, which is the textbook signature of phrase-level memorization at exactly the n-gram length predicted.

Clean-probe collapse on a held-out synthetic test set. On SocratDataset-SYNTHETIC (75 Sonnet-4.6-generated Chinese dialogues, verifiably disjoint from SocratDataset's published splits), SocratTeachLLM's stage-balanced macro state accuracy collapses from $\mathbf{63.40}\%$ on the published test split to $\mathbf{32.86}\%$ on synthetic data; R-1 drops $48.07 \rightarrow 35.72$. On the same synthetic test set, baseline Gemma 4 31B + 10-shot (no SocratDataset fine-tuning) scores stage-balanced 56.13 / R-1 38.76. **SocratTeachLLM under-performs baseline Gemma on truly held-out data, despite outperforming it by $\sim$13 stage-balanced points on the published test split.** The within-model delta (-30.54 pp stage-balanced) is the textbook signature of catastrophic out-of-distribution generalization. Together with the surface-form inversion, the monotonic n-gram widening, and the translation-pipeline reproduction, the most parsimonious account is that SocratTeachLLM was trained on test-set surface forms, either by direct contamination or by insufficient train/test distributional separation.

LLM-judge cross-lingual transfer confirms. On the four-axis Sonnet 4.6 rubric across 15 judged configurations, our locked integration tops the leaderboard at 8.32/10. Cross-lingual judge deltas decide the case: Opus + top-3 transfers Chinese$\rightarrow$English at only -0.07 judge loss, while Sonnet/Opus + SocratTeachLLM lose -1.01 and -1.03 ($14\times$ more) on a paraphrase-invariant metric.

3.4 Sample-size calibration

A bootstrap convergence analysis across all seven of our full-scale n=681 configurations (B=500 subsamples at a 13-point grid) shows the published test split is over-sampled. Cross-configuration plateaus (the smallest n at which 95th-percentile bootstrap deviation from full-n truth stays under ε for every metric and every run, simultaneously) are state-acc 400, R-1 75, R-2 75, BLEU-4 75 at $\varepsilon = 2$ pp. State accuracy is the binding constraint. We adopt $\boldsymbol{n}$**=400** as the canonical ground-truth sample size: it guarantees all four primary metrics within $\leq$ 2 pp at 41% less compute, wall-clock, and API spend than the published n=681.

3.5 Code, datasets, and reproducibility demo

Across the campaign we evaluated **143** model variants (cross-teacher grids, prompt-engineering tournaments, multiple full n=681 runs, frontier-model comparisons, bilingual and contamination probes); the complete master leaderboard, per-run logs, and per-dialogue traces ship in the repository. The locked headline is reproducible end-to-end via the `scripts/eval_bert_gemma_fewshot10_full.sh` entry point, which bootstraps the Gemma 4 31B server, the Qwen3.5-LoRA consultant on CPU, and the n=681 evaluation loop.

- **Code + reproduction scripts:** https://github.com/ulises-c/csen-346
- **Datasets:** SocratDataset (?) mirror, SocratDataset-EN (translation), and SocratDataset-SYNTHETIC (clean-probe contamination control) on Hugging Face under https://huggingface.co/ulises-c.
- **Model:** SocratTeachLLM reproduction mirror at https://huggingface.co/ulises-c/SocratTeachLLM.
- **Slide deck (live demo of the architecture):** `PRESENTATION.md` in the repository (rendered via `reveal-md`); slides include screenshots of per-turn classifier outputs, the integration architecture, and the locked-headline leaderboard.

4 Conclusion and Future Work

Summary of findings. We reproduced KELE on a single-GPU budget and extended it on three axes: a deterministic-classifier consultant integration, a memorization-resistant evaluation panel including the unified score of Eq. 1, and a clean-probe contamination diagnostic against a public synthetic dataset we release. Our locked headline scores $\mathbf{55.39}\%$ state accuracy / unified $\mathbf{72.24}$ at n=681, a $\mathbf{2.14}\times$ lift over the GPT-4o + SocratTeachLLM baseline, with 6.6–9.2$\times$ multipliers on the cognitively hard middle and closure stages, and a $+\mathbf{2.18}$-unified-point lead over the best frontier-teacher configuration we measured on a single 32 GB consumer GPU. Methodologically, we show that surface-form metrics on this benchmark systematically reward training-data memorization over teaching capability; that the published surface-form headline is reproducible by translating the dataset into English; that on truly held-out synthetic data the published baseline collapses below baseline Gemma; and that the published n=681 test split can be replaced by n=400 for 41% compute savings without loss of decision precision. An intermediate *fusion* variant (consultant and teacher emitted by a single JSON-schema-constrained call) yielded an early small-scale lift but its schema-fallback rate proved architecture-dependent at full scale, motivating the deterministic-classifier integration that carries the locked result.

Contributions. (i) Deterministic-classifier consultant integration with a three-step architectural evolution (Chinese BERT $\rightarrow$ Qwen3-Embedding-LoRA $\rightarrow$ Qwen3.5-LoRA), and the architectural thesis that the consultant axis is the binding constraint on canonical-scale performance, validated by an independent matched-classifier negative-result control; (ii) memorization-resistant evaluation panel and the unified score; (iii) four converging contamination diagnostics culminating in the SocratDataset-SYNTHETIC clean-probe; plus open release of code, datasets, classifier checkpoints, and reproduction scripts.

Limitations. The +2.18-point local-overtakes-frontier result is teacher-, prompt-, and consultant-specific (Gemma 4 31B, 10-shot stage-balanced, Qwen3.5-0.8B-LoRA); two adjacent matched-consultant cells (A3B-35B, Qwen-27B no-think) sit 2.25 and 3.35 points behind frontier. The $\leq$ 2-pp n=400 variance budget is comparable to the +2.18 margin, so the claim is well-supported but bounded. Surface-form scores on our open-weight teacher remain $\approx$ 14 R-1 points below the published baseline; we argue this is the wrong primary metric, but readers who weight reproducible phrasing highly should interpret R-1/BLEU-4 as adequacy-on-imitation rather than instructional quality. Primary experiments are in Chinese; the cross-lingual probe via SocratDataset-EN provides a partial English-side check, but full English-domain pedagogical evaluation remains future work.

Future work. (a) The teacher-side Stage 2b QLoRA adapter (Gemma 4 31B, 122 M trainable parameters, final loss 0.4359, merged GGUF Q5_K_M 21 GB) trained stably but its end-to-end evaluation is currently blocked by a train/serve prompt-format mismatch (training data emits multi-turn user/assistant pairs with Pattern-A markers; the inference pipeline emits stringified-history single-turn prompts). The scope-isolated fix (re-prep training data to match the inference shape; $\approx$ 11 GPU-h) is queued. (b) A broader prompt-engineering tournament (10 utilizations $\times$ n=50, then promote top-3 composition to n=681) targets closing the residual R-1 gap. (c) Bilingual co-training of the classifier on the union of SocratDataset and SocratDataset-EN, currently transferable cross-lingually with a -9.24 pp macro drop. (d) A within-stratum stratified n=200–300 sampling protocol may match n=400 random's decision precision at lower cost; we did not measure this. (e) An English-domain Socratic-teaching benchmark constructed natively (not as a translation) would help separate language-bound from method-bound effects in future work.

Ethics Statement

This work uses a publicly released educational dialogue dataset (SocratDataset) collected and licensed by **?** for academic research. No personally identifiable information is processed; no human subjects were involved.

The intended use of the system is automated Socratic-style scaffolding for elementary-school science, deployed as a tutoring aid under teacher supervision rather than as an autonomous instructor. Open-weight teachers occasionally produce factually plausible but incorrect explanations; the state classification we evaluate measures pedagogical strategy selection, not factual correctness of the generated content. Consequential educational deployment should pair the system with ground-truth answer keys and human instructor review of dialogue transcripts. The contamination diagnostic we propose is intended as a methodological tool for evaluating Socratic-teaching benchmarks, not as a judgment about any individual or team behind a published artifact.

All code, configurations, and trained checkpoints are openly released at `https://github.com/ulises-c/csen-346` to enable reproduction, scrutiny, and downstream improvement.

A Classifier Evolution: Per-Stage Detail

Table 6 reports the full per-stage breakdown for the three classifier architectures evaluated in §2 on the held-out 4,304-turn test split. Each column shows implied stage-level accuracy when the 34-state predictions are aggregated up to the parent stage. The progression is concentrated almost entirely on stage c (the 22-way inductive-reasoning classification): the other four stages saturate near ceiling for all three classifiers. The Chinese BERT 5-way head, included for reference, was used only in the early reproduction phase to verify the feasibility of the supervised-classifier approach before the 34-state pipeline classifier was trained.

Stage	BERT (24M)	Qwen3-Emb + LoRA (∼600M)	Qwen3.5 + LoRA (∼800M)
a	100.0	100.0	100.0
b	90.28	90.67	90.93
c	76.16	84.87	**85.14**
d	76.98	74.93	75.89
e	95.45	96.48	96.77
Overall (34-state)	61.64	66.66	**67.57**

Table 6: Per-stage and overall accuracy (%) of the three deterministic-classifier consultants on the 4,304-turn test split. The architectural progression is concentrated on stage c (the binding constraint).

Training compute. BERT (5-way head): 92 s on a single 5090. BERT (34-state head): 148 s. Qwen3-Embedding-0.6B + LoRA: $\approx$ 37 min at batch size 8, with merged-weight checkpoint at $\approx$ 2.3 GB. Qwen3.5-0.8B + LoRA r=8: $\approx$ 50 min at batch size 16 with bf16 autocast and gradient checkpointing, after a small hyperparameter sweep (bs $\in \{4, 16, 32, 48\}$). Per-state breakdowns and full confusion matrices for all three classifiers are shipped in the repository at `results/state-clf-*/test_eval.json`.

B LLM-Judge Four-Axis Breakdown

The unified score's judge component (§2, Eq. 1) decomposes into four memorization-resistant axes scored independently by Claude Sonnet 4.6: *Socratic validity* (max 3), *learning advancement* (max 3), *age-appropriateness* (max 2), and *question-form fidelity* (max 2). Table 7 reports the per-axis breakdown for the locked headline plus three reference cells.

Per-stage judge. For the locked headline at n=681, per-stage judge averages are a=9.38, b=8.82, c=8.32, d=7.43, e=7.14. Closure (stage e) is the most demanding axis for the LLM judge across all configurations, reflecting the difficulty of producing pedagogically tight summarization rather than a routing failure.

Configuration (n=681)	Val. (/3)	Adv. (/3)	Age (/2)	Q-form (/2)	Total (/10)
Locked: Qwen3.5-LoRA + Gemma	**2.34**	**2.52**	1.97	1.48	**8.32**
BERT-fixed + Gemma	2.30	2.50	1.97	1.48	8.25
Legacy BERT + Gemma	2.27	2.49	1.97	1.46	8.19
BERT + Claude-Sonnet + top-3	2.27	2.48	1.98	1.46	8.19

Table 7: Per-axis judge scores for the locked headline and three control cells at n=681. All cells score nearly identically on the age-appropriateness and question-form axes (the teacher's surface behavior is consistent across consultant swaps). The headline's advantage concentrates on Socratic validity and learning advancement, the two axes that reward correct pedagogical routing.

Cost. The full memorization-resistant judge sweep across all 15 judged configurations cost $59.86 in total Anthropic API spend; each n=681 judge pass is approximately $15.30 at 16 parallel workers, completing in $\approx$ 12 minutes.

C Cross-Lingual Translation and Clean-Probe Full Data

Table 8 extends the cross-lingual translation finding from §3.3: running SocratTeachLLM as teacher against the SocratDataset-EN translation with frontier Claude consultants produces surface-form scores nearly identical to the published Chinese headline. The R-2/BLEU-4 ratio jumps from $\approx$ 1.3 (Chinese) to $\approx$ 9 (English) under the identical teacher, isolating phrase-level memorization at the 4-gram length.

Configuration	R-1	R-2	BLEU-4	R-2/B-4
Peng et al. (2025) STL headline (ZH)	57.40	33.63	41.96	$\approx$ 1.3
Sonnet + STL (ZH, clean rerun)	45.61	—	—	$\approx$ 1.3
Sonnet + STL (EN)	**55.85**	**33.79**	3.56	$\approx$ **9**
Opus + STL (EN)	44.22	26.20	2.96	$\approx$ 9

Table 8: Cross-lingual translation experiment. Sonnet + SocratTeachLLM on SocratDataset-EN reproduces the published headline of R-1 = 57.40, R-2 = 33.63 within 1.55 R-1 points and 0.16 R-2 points; the R-2/B-4 ratio simultaneously inflates to $\approx$ 9, isolating the loss of Chinese 4-gram fingerprints under translation.

Clean-probe details. SocratDataset-SYNTHETIC contains 75 Chinese Socratic dialogues generated end-to-end by Claude Sonnet 4.6 on the same elementary-school science topic distribution as SocratDataset. The generation prompt enforces the SocRule 5-stage progression and 34-state label set; we ran the generated dialogues through a SocRule-compliance validator before release. SocratTeachLLM scored stage-balanced macro 63.40% / R-1 48.07 on the published test split versus 32.86% / 35.72 on SocratDataset-SYNTHETIC, an absolute drop of 30.54 pp stage-balanced and 12.35 R-1. Baseline Gemma 4 31B + 10-shot (no SocratDataset fine-tuning) scored 56.13 / 38.76 on the same synthetic test set: an upper bound on what a non-contaminated model achieves on the clean-probe distribution.

D Bootstrap Convergence Analysis

§3 reports our n=400 canonical-sample-size recommendation derived from a bootstrap convergence analysis across all seven of our full-scale n=681 configurations. Table 9 shows the cross-configuration plateau (the smallest n at which the 95th-percentile bootstrap deviation from the full-n truth stays under tolerance ε, simultaneously for every metric and every configuration). Figure 2 plots the convergence curves themselves; the state-accuracy curve dominates the convergence rate.

ε (pp)	state-acc	R-1	R-2	sent-BLEU-4
0.50	600	500	500	400
1.00	600	200	200	175
2.00	**400**	75	75	75

Table 9: Cross-configuration convergence plateaus across all seven full-scale runs in our campaign. The recommended canonical n=400 holds all four metrics within $\leq$ 2 pp of the full-n truth.

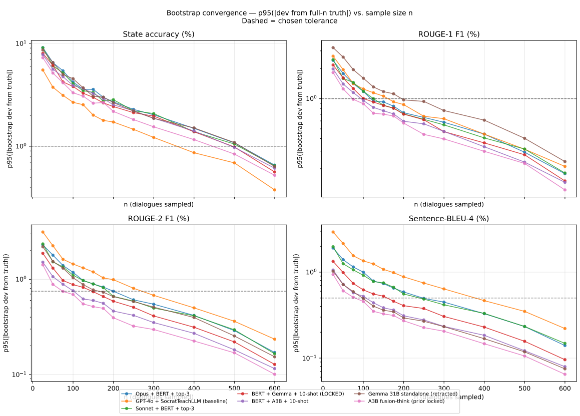

Figure 2: Bootstrap convergence curves (B=500 subsamples without replacement) at a 13-point grid from n=25 to n=600 for state accuracy, ROUGE-1, ROUGE-2, and sentence-BLEU-4 across our seven n=681 configurations. State accuracy converges 3–4$\times$ slower than the surface-form metrics because the per-turn 34-class indicator has higher per-dialogue variance than the continuous, tightly-distributed surface-form F1s.

Stratified-sampling extension. A stage-stratified subsample (balanced over stage and topic) may reach the same $\leq$ 2-pp precision at n=200–300, but we did not measure this; it is flagged as future work in §4.

E Schema-Fallback Analysis at Full Scale

The fusion-architecture retraction described in §2 rests on a single concrete failure: Gemma 4 31B's strict-JSON schema-fallback rate at full scale.

Backbone	Turns	Fallback turns	Rate
Qwen 35B-A3B (fusion)	4,171	38	0.91%
Gemma 4 31B (fusion)	4,246	890	**21.0%**
Integration (any teacher)	$\geq$ 3,762	0 (by construction)	0.0%

Table 10: Schema-fallback rate at full n=681 scale by architecture. Qwen 35B-A3B handles strict-JSON grammar within tolerance; Gemma 4 31B in fusion mode collapses on 21% of turns. The deterministic-classifier integration removes the JSON-schema constraint from the consultant path entirely, structurally bypassing the failure mode.

Predictive value of schema-fallback rate. The Gemma standalone smoke–mini-average prediction (§2) was 46.71% state accuracy; the realized full-run number was 31.39%, a 15.32-pt overshoot. The mini sample (n=25 dialogues) showed 0/148 fallback turns and did not surface the issue. We treat the lesson as methodological: *cross-architecture scaling-prediction protocols must triangulate schema-fallback rates at full scale*, not just at the mini-tier sample.

F Stage 2b QLoRA SFT Training Details

Future-work item (a) in §4 references a Stage 2b QLoRA adapter on Gemma 4 31B that trained successfully but whose end-to-end evaluation is currently blocked by a train/serve prompt-format mismatch. Training-side details are summarized in Table 11; loss descended monotonically from 0.65 to 0.44 over 2,298 steps with no spikes, no divergence, and a final mean-token-accuracy of 0.86.

Train/serve schema drift (the eval-blocking failure). The training data pipeline emits multi-turn user/assistant message pairs whose user role carries the raw student utterance plus two Chinese-language "Pattern-A" marker prefixes (English gloss: *"Socratic-consultant evaluation result"* and *"Socratic-consultant suggested action"*).

Setting	**Value**
Base model	`google/gemma-4-31b-it`
Method	QLoRA 4-bit NF4, r=16, α=32
Targets	q/k/v/o + gate/up/down proj.
Learning rate	5×10^{-5}
Batch / grad-accum	$1 \times 16 = 16$
Sequence length	1024 (forced from 1280 by VRAM)
Epochs / steps	3 / 2,298
Trainable params	122 M (0.39% of 31.4 B)
Final train loss	0.4359
Final tok-acc	0.8555
Wall clock	10 h 49 m on a single 5090
Adapter size	468 MB safetensors
Merged GGUF (Q5_K_M)	21 GB

Table 11: Stage 2b QLoRA SFT training configuration and outcomes.

The inference pipeline emits a single-turn message with stringified dialogue history under two different Chinese-language prefixes (gloss: *"Dialogue history"* and *"Current student input"*) that the SFT model never saw during training. On the same checkpoint and server, training-format input produces a clean Socratic question (20 tokens, finish_reason=stop, 0.5 s); inference-format input produces a 2,048-token repetition loop (finish_reason=length). The model is healthy; the eval pipeline is sending it out-of-distribution prompts.

Fix paths. Option A (preferred, scope-isolated): re-prep the training data such that each user message matches the inference pipeline's stringified-history shape verbatim, then re-run Stage 2b with identical hyperparameters ($\approx$ 11 GPU-h on the 5090, no other code change). **Option B:** modify the inference pipeline to emit alternating user/assistant message pairs, but this would shift the prompts seen by every other teacher LLM in the evaluation suite and require re-baselining the locked-headline-adjacent cells. We prefer Option A. The training artifacts (adapter, merged GGUF) and the merge+quantize pipeline ship in the repository so a future session can land the eval without re-deriving infrastructure.